A different series in the same calculus:
the Edgeworth expansion of a U-statistic as connected-diagram Einsums

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Abstract

The growth-rate series $Z_L = \sum_{|I|=L} \omega(I)$ collapses, when its dependency graph is a chain, to a transfer matrix – there the EINSUM/treewidth calculus is just vocabulary. Here is a series of a genuinely different shape, in the companion paper's own domain, where the calculus does real work: the *Edgeworth expansion* of a U-statistic. It is an asymptotic series in $n^{-1/2}$ for the distribution of the statistic, and its coefficients are the *cumulants*, which are sums over *connected diagrams* – each an EINSUM of the kernel's components, with the diagram's treewidth setting its cost. The moment↔cumulant map is Möbius inversion on the partition lattice, the same inclusion–exclusion as the paper's $U \leftrightarrow V$ relation. We make the leading coefficient explicit (it is a sum of two diagrams, and a single diagram gives the wrong answer – here even the wrong sign), verify it against exact cumulants and Monte Carlo, and show the expansion approaching normality at the predicted rates.

1 A series that is not a growth rate

Let $X_1, \ldots, X_n$ be i.i.d. and let U_n be a U-statistic of order m with symmetric kernel h,

$$U_n = \binom{n}{m}^{-1} \sum_{i_1 < \cdots < i_m} h(X_{i_1}, \ldots, X_{i_m}), \qquad \theta = \mathbb{E}[U_n]. \tag{1}$$

The companion paper computes such objects exactly by reading U_n (or its V-statistic cousin) as an EINSUM $\sum_\alpha \prod_k T_k(\alpha[A_k])$. Here we ask a different question – not the *value* of U_n but the *shape of its distribution*. Standardise $W_n = (U_n - \theta)/\sigma_n$, $\sigma_n^2 = \mathrm{Var}(U_n)$. The central limit theorem says $W_n \Rightarrow \mathcal{N}(0,1)$; the *Edgeworth expansion* is the refinement[1]

$$\boxed{P(W_n \le x) = \Phi(x) - \phi(x)\left[\frac{\lambda_3}{6} He_2(x) + \frac{\lambda_4}{24} He_3(x) + \frac{\lambda_3^2}{72} He_5(x) + \cdots\right]}, \tag{2}$$

where ϕ, Φ are the standard normal density and c.d.f., He_k are the Hermite polynomials, and $\lambda_r = \mathrm{cum}_r(W_n)$ is the r-th *standardised cumulant*, of size $\lambda_r = O\big(n^{-(r-2)/2}\big)$. This is the new kind of series: an *asymptotic* expansion in powers of $n^{-1/2}$, indexed by cumulant order rather than by word length. "Convergence" here is not a growth rate but the *rate of approach to normality*: truncating after the k-th term leaves an error $O\big(n^{-(k+1)/2}\big)$ (under Cramér's condition), and the whole series is typically divergent, valid up to an optimal truncation.

2 The coefficients are cumulants, and cumulants are diagrams

The content of (2) is entirely in its coefficients λ_r. Two facts place them squarely in the companion paper's calculus.

[1] For U-statistics: R. Helmers and W. van Zwet; M. Callaert, P. Janssen and N. Veraverbeke, *Ann. Statist.* (1980); P. Bickel, F. Götze and W. van Zwet (1986). For sums it is classical (Cramér, Esseen).

Cumulants = Möbius inversion of moments. The map between moments and cumulants is exactly Möbius inversion over the lattice of set partitions,[2]

$$\mathrm{cum}_r(Y_1,\dots,Y_r) = \sum_{\pi\in\Pi_r} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B\in\pi} \mathbb{E}\Big[\textstyle\prod_{i\in B} Y_i\Big]. \tag{3}$$

This is the *same* inclusion–exclusion that turns a V-statistic (all index tuples) into a U-statistic (distinct indices) in the companion paper: cumulants are the "connected" (distinct-block) part of the moments.

Each diagram is an Einsum. Expand U_n in its Hoeffding components $g_1, g_2, \dots$ (the canonical projections of h). For a finite alphabet $\{1,\dots,q\}$ with law p – so the kernel is a tensor H and everything is a finite contraction – the order-2 pieces are

$$\theta = p^\top Hp, \quad g_1 = Hp - \theta\cdot\mathbf{1}, \quad g_2[a,b] = H[a,b] - g_1[a] - g_1[b] - \theta, \quad \zeta_1 = \textstyle\sum_x p_x g_1(x)^2. \tag{4}$$

Each cumulant λ_r is then a sum over *connected diagrams*: vertices are sampled indices, a g_1 sits on one index and a g_2 couples two, and the value of a diagram is the EINSUM that contracts these components against p. The diagram's *treewidth* is the cost of that contraction – the companion paper's bound (4) verbatim. For order-2 kernels the diagrams are small; for order-m kernels and higher cumulants they are genuinely non-chain graphs, and that is where the treewidth calculus earns its keep.

3 The leading coefficient: a sum of two diagrams

The skewness coefficient λ_3 is the first non-Gaussian term. To leading order in n it is *not* a single diagram – it is a sum of two connected ones (Figure 1):

$$\lambda_3 = \frac{c_3}{\sqrt{n}} + O(n^{-3/2}), \qquad \boxed{c_3 = \frac{\gamma_1 + 3J}{\zeta_1^{3/2}},} \qquad \gamma_1 = \sum_x p_x\, g_1(x)^3, \quad J = \sum_{a,b} p_a p_b\, g_1(a) g_1(b)\, g_2(a,b). \tag{5}$$

γ_1 is the "g_1^3" diagram (three projections on one index); J is the "$g_1 - g_1 - g_2$" cross diagram (two indices joined by a g_2). The lesson is sharp: keeping only the Hájek/projection diagram γ_1 is *wrong*, because the cross diagram cancels most of it.

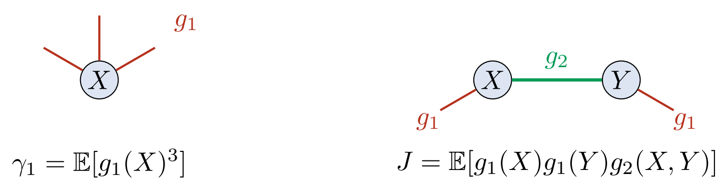

Figure 1: The two connected diagrams in the leading skewness (5). Each is an EINSUM contracting the Hoeffding components g_1, g_2 against the law p. Higher cumulants and higher-order kernels give larger, non-chain diagrams whose treewidth sets the cost.

A concrete check. Take the 5-letter kernel and law in `code/exp7_edgeworth.py`. The EINSUMs give $\gamma_1 = -0.0140$, $J = +0.0196$, $\zeta_1 = 0.128$, hence

$$c_3 = \frac{\gamma_1 + 3J}{\zeta_1^{3/2}} = 0.977, \qquad \text{whereas the single diagram} \quad \frac{\gamma_1}{\zeta_1^{3/2}} = -0.305$$

has the *wrong sign*. Computing the *exact* cumulants of U_n (via the multinomial factorial moments, no asymptotics) confirms $\lambda_3\sqrt{n} \to 0.977$ as n grows $(0.54, 0.72, 0.84, 0.91, 0.94$ at

[2]T. P. Speed, *Cumulants and partition lattices*, Austral. J. Statist. (1983); G.-C. Rota.

$n = 16, 32, 64, 128, 256$), and a 1.6×10^7-draw Monte-Carlo gives skewness 0.092 against the exact $\lambda_3 = 0.105$ at $n = 64$. The two-diagram Einsum formula (5) is correct; the one-diagram shortcut is not.

4 The rate of the expansion

Because $\lambda_r = O(n^{-(r-2)/2})$, the successive terms of (2) shrink by a factor $n^{-1/2}$ each: the normal approximation errs at $O(n^{-1/2})$ (the skewness term), adding it gives $O(n^{-1})$, adding the kurtosis term gives $O(n^{-3/2})$, and so on. Figure 2 demonstrates both halves: the Edgeworth density captures the skew the normal misses, and each extra term lowers the sup-distance to the true law at the predicted rate.

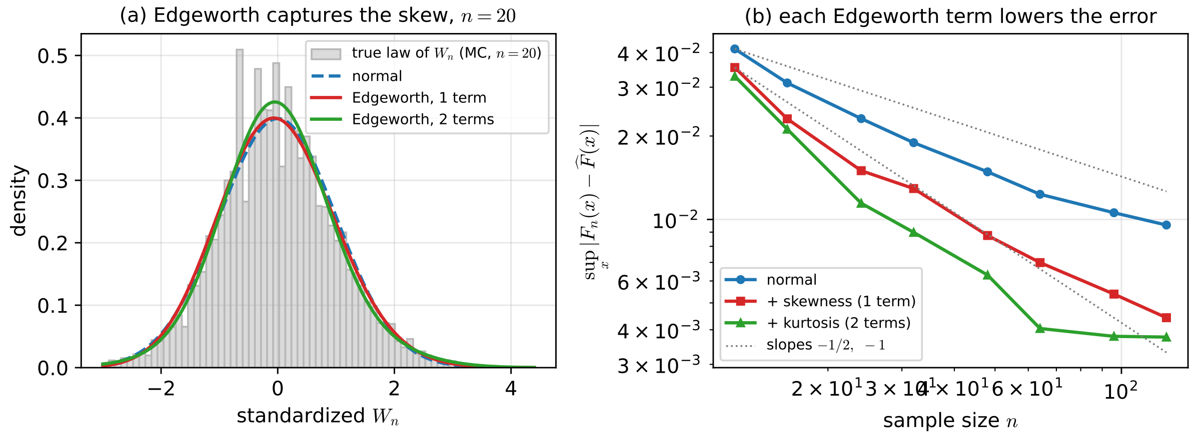

Figure 2: Edgeworth expansion of a U-statistic. (a) at $n = 20$ the one- and two-term expansions track the (right-skewed) true law that the normal cannot. (b) the sup-error $\sup_x |F_n - \widehat{F}|$ against a 1.6×10^7-draw Monte-Carlo truth: each Edgeworth term lowers it, along slopes near $-\frac{1}{2}$ and -1. (Discreteness of a small-alphabet statistic caps the gain at small n; the coefficients themselves are exact.)

Observation 1 (what is being computed, and why it is the same calculus)**.** The Edgeworth coefficients are cumulants; cumulants are connected sums of diagrams; each diagram is an Einsum of the kernel's Hoeffding components; and the moment→cumulant passage is the partition-lattice Möbius inversion that the companion paper already uses for $U \to V$. The treewidth of a diagram is the cost of its contraction. So computing high-order distributional corrections for high-order U-statistics *is* the paper's treewidth-bounded Einsum problem, now indexed by cumulant order.

5 Where treewidth bites, and what is next

Higher order is where it is non-trivial. For an order-2 kernel the leading diagrams are a point and an edge. For an order-m kernel, the r-th cumulant is a sum over connected diagrams on r kernel-copies sharing arguments; these are general graphs, the optimal contraction is the NP-hard treewidth problem, and the heuristics of `opt_einsum`/`cotengra` are exactly the tool for enumerating them.[3] This is the genuine, non-chain home of the calculus – unlike the transfer-matrix series, nothing here collapses to a single quadratic form.

[3] D. G. A. Smith and J. Gray (2018); J. Gray and S. Kourtis (2021). The diagram/graph formalism for cumulants: P. Major, *Multiple Wiener–Itô Integrals*; S. Janson, *Gaussian Hilbert Spaces*.

Relatives in the same family. The same coefficients drive the *Cornish–Fisher* inversion (quantiles), the *saddlepoint* approximation (an exponential reordering of the same cumulants), and the second-order correctness of the *bootstrap*. The *degenerate* case ($\zeta_1 = 0$, e.g. $\mathbb{E}[h(x, X)] \equiv \theta$) is a different regime entirely: nU_n converges to a weighted χ^2 via the spectrum of the operator g_2 – an eigenvalue series rather than a normal one, and a natural next target.

Honest caveats. The series is asymptotic, not convergent; the diagram count grows fast with order; for lattice statistics a separate oscillatory correction is needed (visible as the small-n floor in Figure 2b); and treewidth fixes the *cost* of a coefficient, not its value.

Reproduce: `python3 code/exp7_edgeworth.py` computes the diagram EINSUMs, the exact cumulants, and the figure; it prints the verification of (5).